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ORIGINAL PAPER

Multi-criteria fuzzy decision support for conceptual evaluation in design of mechatronic systems: a quadrotor design case study

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Abstract
Designing mechatronic systems is known to be a very complex and tedious process due to the high number of system components, their multi-physical aspects, the couplings between the different domains involved in the product, and the interacting design objectives. This inherent complexity calls for the crucial need of a systematic and multi-objective design thinking methodology to replace the often-used sequential design approach that tends to deal with the different domains and their corresponding design objectives separately leading to functional but not necessarily optimal designs. Thus, a new approach based on a multi-criteria profile for mechatronic systems is presented in this paper for the conceptual design stage. Additionally, to facilitate fitting the intuitive requirements for decision-making in the presence of interacting criteria, three different methods are proposed and compared using a case study of designing a vision-guided quadrotor drone system. These methods benefit from three different aggregation techniques such as Choquet integral, Sugeno integral and fuzzy-based neural network. To validate the decision yielded by the results of global concept score for each aggregation methods, a computer simulation of a visual servoing system on all design alternatives for quadrotor drone has been performed. It is shown that although the Sugeno fuzzy can be a useful aggregation function for decisions under uncertainty, but the approaches using Choquet fuzzy and fuzzy integral-based neural network seem to be more precise and reliable in a multi-criteria design problem where interaction between the objectives cannot be overlooked.

Keywords Decision support · Mechatronic systems · Multi-criteria design · Fuzzy logic · Quadrotor system

1 Introduction

Introduction of mechatronic systems was a direct result of an increased need for controlled and intelligent electromechanical systems with better performance, more flexibility and higher reliability capable of performing complex tasks. The process of designing a wide variety of products such as avionic systems, robots, transportation and construction machines or even home appliances is now considered within the area of mechatronic systems design that requires coherent design activities between several disciplines such as mechanical, electrical, control and software engineering. The high number of their components, their multi-physical aspect and the couplings between the different disciplines involved make the design task very tedious and complex which requires a significant amount of time (Torry-Smith et al. 2011, 2012; Mohebbi et al. 2014). Therefore, this inherent complexity and the dynamic couplings between subsystems, urges for a methodology to find the optimal design solutions and to replace the often-used sequential design approach that tends to deal with the different subsystems and domains (i.e., mechanical, electrical, software, fluid, thermal, etc.) separately. Moreover, a systematic and multi-objective design methodology is needed. The resulting products would eventually form a spatial integration and a functional interaction in components, modules, products and systems.

Design activities, including of mechatronic products, include three major phases: conceptual design, detailed design, and prototyping and improvements. Conceptual

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design is an early stage of design in which concepts are selected and employed to solve a given design problem and then a decision is made on how to interconnect these concepts into an appropriate system architecture (Rzevski 2003). Moreover, at this stage, a comprehensive and consistent listing of the requirements and behaviors is required as well as a thorough identification of critical parts of the solution that affects the overall performance. Rzevski (2003) discusses conceptual design of mechatronic systems based on multi-agent technology. He argues that the best configuration of a design process is a network of autonomous, intelligent decision-making units able to reach decisions through the process of negotiation. However, the given examples of systems designed by such methodology indicated the use of high number of sensory components and control agents as well as geometrically variable parts, which all together could elevate the cost of production and increase the risk of failure. Ziv-Av and Reich (2005) presented a comprehensive approach for generating optimal concepts in diverse disciplines. They suggested decomposing a complex problem into smaller subproblems, to use highly simplified evaluations. In another effort (Reich and Ziv Av 2005), they defined a design robustness term as the stability of the optimal concept or configuration generated by the approach presented in (Ziv-Av and Reich 2005) with respect to variations in many factors such as designers' subjective judgment, availability of the technology, organization context, and the preferences of the customers.

A number of problems and limitations are encountered when design is at its early stages, as it requires the selection of components and choosing between alternatives for software and control strategies (Rzevski 2003; Coelingh et al. 2002; Moulianitis et al. 2004). This part of conceptual and preliminary design is usually known as choosing the "Elite Set". This practice creates challenges due to insufficient support of the multi-criteria nature of mechatronic system design, which calls for decision support across various disciplines. In such cases, engineers tend to choose the first and the best components from what they see as available and feasible to meet their design requirements. Such decisions can often lead to a functional design, but rarely to an optimal one. This ill decision-making generally occurs due to improperly defined performance criteria and lack of knowledge about the co-influences between criteria and the functionality to be provided by neighboring disciplines. Avigad and Moshaiov (2009) presented an approach to support the selection of conceptual solutions for multi-objective problems. Their method involved performing a set-based comparison between concepts by considering both optimality and variability. Their method was consistent with the so-called Toyota set-based concurrent engineering process. Moulianitis et al. (2004) proposed a methodology for decision-making in conceptual mechatronic design based upon an evaluation index including three criteria: intelligence, flexibility, and complexity. Weight factors were manually applied to highlight the importance of each criterion. The formulation of the evaluation score has been presented based on t-norms (Klement et al. 2000) and averaging operators. However, the methodology does not consider interactions between criteria, and a limited, discrete search space was considered. A producibility index vector has been proposed by Byun and Elsayed (1996), in which both quality and cost requirements are considered. Their producibility index is defined as a measure of the desirability of a product, quality of process design and manufacturing costs. Based on this index, overall process capability is calculated using weighted geometric means. The main drawback of their work is that the design evaluation for various alternatives is achieved by direct comparison of every element of the index vector.

De Silva (2004) stated that in a sequential design approach for mechatronic systems, optimal design of subsystems does not necessarily provide the optimum overall configuration. He proposed to associate performance indices to the mechatronic subsystems within an indicator, called "mechatronic design quotient (MDQ)" and maximizing this indicator after integrating all the subsystems. Based on this method, Behbahani and Silva (2007) proposed a formal and systematic framework for design of a mechatronic system using the concepts of mechatronic design quotient (MDQ) in a concurrent design approach, where correlations between design criteria have been taken into account using fuzzy concepts to define them. MDQ was implemented in pilot projects (Behbahani and Silva 2008), and was claimed to be efficient; however, measurement and determination of criteria for design are very qualitative, and no systematic measurement approach has been presented nor implemented. In a different approach, Janschek (2012) proposed a method based on performance metrics defined as the deviation of the measured criteria relative to a reference value. *P*-norm of the standard deviations of the p criteria was used to define the standard deviation of the overall mechatronic design. Correlations and interactions between objectives were also taken into account by calculating the covariance values between different criteria. For the same purpose of performance analysis in conceptual design, Ferreira and Gil (2012) proposed using feedforward multi-layer perceptron neural networks as a decision support to recommend design solutions for engineering systems based on existing database of successful designs. They focused on reducing the complexity of the neural model from the designer's point of view. To measure the quality of the design prediction and to avoid misleading solutions an error index was also provided.

It can be understood that although a number of efforts focused on presenting supports for decision-making in design, there is still a need for a comprehensive approach that simultaneously considers a set of general main criteria

for conceptual design of mechatronic products, and also provides the necessary means to assess these criteria and aids the designers to decide among all generated alternatives. As the main contribution, this paper presents a systematic approach for supporting multi-objective and concurrent design of mechatronic systems in early stages of design. First and after a thorough literature study and analysis of the available research work on mechatronic system design, a vector of five main criteria and corresponding sub-criteria for conceptual design are identified and then the assessment methods for each one of them are explained and in some cases adapted to the case of designing a complex mechatronic system. This provides a unified framework for evaluation of design concepts for mechatronic systems. Consequently, in Sect. 2, the new index vector—mechatronic multi-criteria profile (MMP), is introduced for the purpose of concept evaluation in mechatronic systems design. MMP can be determined numerically through mathematical measures by including sub-criteria and by reflecting an aggregated value for each main criterion. Section 3 is devoted to presenting and comparing three different methods for decision support and aggregation based on fuzzy sets in the presence of interacting criteria by incorporating Choquet integrals, Sugeno fuzzy and fuzzy-based neural networks. Finally, in Sect. 4, the effectiveness of the proposed approach alongside the presented decision support methods is studied and compared by applying it to the conceptual design of a vision-guided quadrotor UAV.

2 Multi-criteria profile (MMP)

One of the most challenging problems faced with regards to design of mechatronic systems is to find the right set of criteria to concurrently evaluate and synthesize the designs. Making design decisions with multiple criteria, in general and for mechatronic systems in particular, is often performed using a Pareto approach. Achieving optimal solutions is a very complex if not impossible task without the identification of the performance parameters involved and the understanding and taking into account their co-influences. Optimal mechatronic design simultaneously requires a precise and systematic design evaluation stage. The lack of simultaneous consideration of objectives within various domains involved at the early design stage increases the need for iterations. This evaluation includes both comparison and decision-making (Ullman 2003). In other words, decision-making is achieved by selecting the "best" alternative by comparison. It is crucial to take into account both correlation between system requirements and also interactions between the multi-disciplinary subsystems (Mohebbi et al. 2014). The goal of concept evaluation is to compare the generated concepts against the requirements and to select the best one for the detailed design and optimization stages. In a conceptual design stage and based on sets of design specifications and goals, candidate solutions are generated (Fig. 1). In most of the design projects, more than one candidate solution is generated and evaluated to select the best one meeting the design objectives and constraints.

To form an integrated and systematic evaluation stage, the most important quantitative criteria have been identified to form an index vector of five normalized elements called mechatronic multi-criteria profile (MMP) as follows:

$$\mathrm{MMP} = [\mathrm{MIQ\ RS\ CX\ FX\ CT}]^T, \tag{1}$$

where MIQ is the machine intelligence quotient, RS is the reliability score, CX is the complexity, FX is the flexibility and CT is the cost of manufacture and production. We also define m_i as the values for the members of MMP sorted in ascending order such that $m_1 \leq m_2 \leq \cdots \leq m_5$ and $0 \leq m_i \leq 1$.

2.1 Machine intelligence

The term "Machine intelligence quotient (MIQ)" was first introduced by Bien et al. (2002) and Kim and Kim (1998) as an index used to assess the intelligence of a control system in mechatronic machines. This index significantly differs from other well-known indices such as control performance, reliability, fault diagnosis capability, etc. Bien et al. (2002) also proposed two measurement methods: the ontological method and the phenomenological method. In Kim's work (Kim and Kim 1998), control performance, fault diagnosis capability

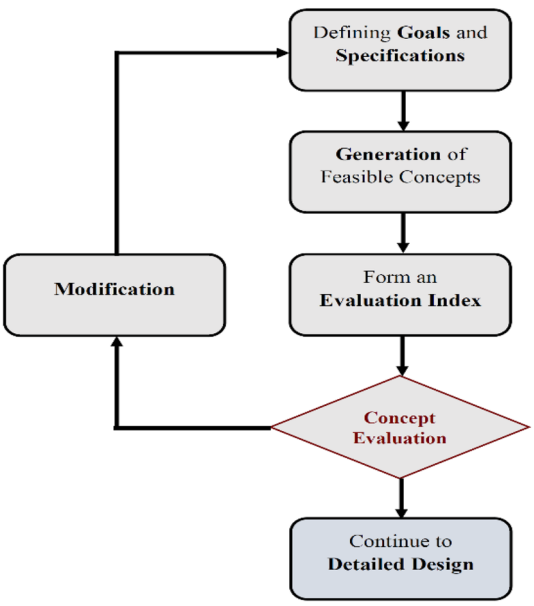


Fig. 1 The usage of concept evaluation in the overall conceptual design process, adopted from (Moulianitis et al. 2004)

and reliability were defined as the main factors in machine intelligence. The term MIQ used in the newly introduced MMP will be assessed based on a method presented by Park et al. (2001) in which the machine intelligence is divided into two components of control intelligence and interface intelligence. Accordingly, the mechatronic system was modeled using an intelligent task graph (ITG). In the topics related to parallel processing and scheduling, the set of tasks and their data dependencies are usually described by data flow graph (DFG). They modified the DFG from the viewpoint of intelligence and transformed it into the ITG, which is suitable for analyzing the machine intelligence. Furthermore, the intelligence required for interaction between the human operator and the machine is taken into account.

Some examples were given by Park et al. (2001) for assessing expert systems for power plants to verify the proposed modeling and measuring procedures. Studying their examples shows that their approach has been more or less aligned with that specific line of application which cannot necessarily contribute to a complex robotic mechatronic system. A number of research works have been dedicated to further adapt this method for specific applications (Anthony and Jannett 2007; Behbahani 2007).

Here, we provide a modified approach to model and assess the MIQ for a robotic platform and at the end in Sect. 4.1, we apply the method to a visual servoing system for quadrotor UAV. Figure 2 shows an example of the ITG in which the circles denote the tasks of control jobs and the directional arrows denote information flow from one task to another. With regards to an ITG, a task set $T = \{T_1, T_2, \ldots, T_n\}$ is the set of n tasks required to control events. $\tau = \{\tau_1, \tau_2, \ldots, \tau_n\}$ is a set of task intelligence costs, in which, τ_i is a scalar

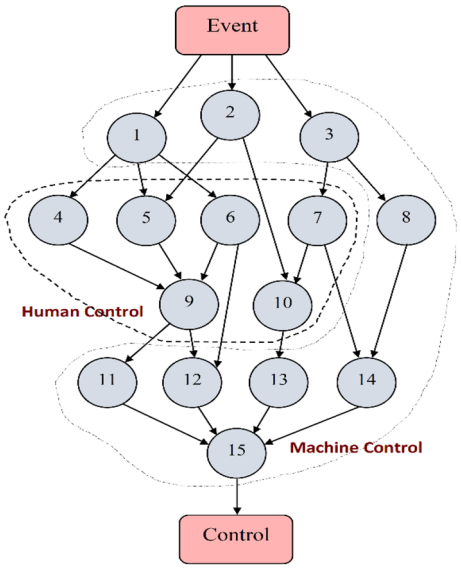


Fig. 2 Example of intelligent task graph (ITG)

representing a quantified measure of intelligence which is required to execute T_i. Accordingly, in a data transfer matrix F, f_{ij} is a scalar showing the data quantity transferred from task T_i to T_j such that:

$$F = \begin{pmatrix} 0 & f_{12} & f_{13} & \cdots & f_{1n} \\ f_{21} & 0 & f_{23} & \cdots & f_{2n} \\ f_{31} & f_{32} & 0 & \cdots & f_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f_{n1} & f_{n2} & f_{n3} & \cdots & 0 \end{pmatrix}. \tag{2}$$

The determination process for the elements of the data transfer matrix from Eq. (2) is explained in a case study in Sect. 4.1.1.

Interface complexity is the complexity of transferring one unit of data between the operator and the machine using user interface devices such as displays, keyboards, control switches, and haptic devices. c_{hm} is defined as the interface complexity of transferring data from the operator (user) to the machine. Accordingly, c_{mh} is defined as the interface complexity of transferring data from the machine to the user. Average interface complexity of a component-based system is calculated as follows:

$$c = \frac{1}{n} \sum_{i=1}^{n} \left(N_i + O_i\right), \tag{3}$$

where n is the number of all components and interface devices in the system and N_i and O_i are the number of incoming and outgoing interactions for every component. Interface intelligence cost $f_{ij}.c_{\text{hm}}$ and $f_{ij}.c_{\text{mh}}$ are the user intelligence amount required to communicate data with the machine, which is proportional to both data quantity and interface complexity.

In a task allocation matrix, A, a_{i1} is equal to 1, if the machine performs task T_i, and a_{i2} is equal to 1 if the human performs task T_i. If the task T_i is shared between the machine and the human user, or, it is not trivial to identify who will be responsible for it, a_{i3} is equal to 1. Thus,

$$a_{i1} + a_{i2} + a_{i3} = 1. \tag{4}$$

For $\forall i,\ 1 \le i \le n$, the n by 3 matrix A is defined as

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} \end{pmatrix}. \tag{5}$$

The control intelligence quotient or CIQ is calculated as the sum of all task intelligence costs,

$$\text{CIQ} = \sum_{i=1}^{n} \left(a_{i1}\tau_i + a_{i2}\tau_i\right). \tag{6}$$

The human intelligence quotient (HIQ) is defined as the required intelligence quantity from the human for controlling plants and is calculated such that

$$\mathrm{HIQ} = \left(\sum_{i=1}^{n} a_{i2}\tau_i\right) + \left(c_{\mathrm{mh}} \sum_{i=1}^{n}\sum_{j=1}^{n} a_{i1}a_{j2}f_{ij}\right) + \left(c_{\mathrm{hm}} \sum_{i=1}^{n}\sum_{j=1}^{n} a_{i2}a_{j1}f_{ij}\right). \tag{7}$$

Finally, The MIQ can be calculated as follows:

$$\mathrm{MIQ} = \mathrm{CIQ} - \mathrm{HIQ}. \tag{8}$$

2.2 System reliability

System reliability assessment (SRA) has been addressed as an important issue of the design process of mechatronic systems in many research works (Behbahani 2007; Silva 2004). Reliability engineering mainly deals with analyzing the expected or actual reliability of a product, process or service, and identifying the actions to reduce failures or their malicious effects. To achieve this goal, all levels of design and production require a process of reliability engineering. A number of methods for reliability assessment have been developed, most of which estimate the system reliability using only the data of components (Zhong et al. 2010). The methods of fault tree analysis (FTA) and failure modes and effect analysis (FMEA) are the most popular tools for reliability assessment, while some other research have been established based upon using Petri nets (PN) for modeling of dynamic characteristics of complex mechatronic systems (Behbahani 2007). To avoid complexity, the dynamic effects for reliability assessment can be neglected and for a multi-components system with no redundancy, the global reliability score (RS) can be expressed by

$$\mathrm{RS} = \prod_{i=1} \left(1 - p_i\right), \tag{9}$$

in which p_i is the failure probability of ith component and m is the number of components in the mechatronic design concept.

2.3 Design complexity

Due to the multi-disciplinary nature of mechatronic systems, the complexity of products themselves and also the complexity of product design and development are significantly increased. Despite design complexity being considered as one of the most important challenges faced by mechatronic designers, just a few work efforts have been carried out for quantitative assessment and estimation of complexity in a design procedure for mechatronic systems (Bashir and Thomson 1999). In the presented paper, the term "Complexity", which was introduced earlier as a member of MMP, is defined by modifying and expanding the work presented in (Moulianitis et al. 2004). The sources of complexity are identified and represented as a vector of six elements as follows:

$$\mathbf{\Phi} = \left[\varphi_1,\ \varphi_2,\ \varphi_3,\ \varphi_4,\ \varphi_5,\ \varphi_6\right]^T, \tag{10}$$

where φ_1 is the number of components as the first and most common source of complexity, φ_2 is the degree of architecture complexity (number of interconnections), φ_3 is the design com plexity which is specified by the backtracks to the earlier stages (number of feedback loops in design process), φ_4 is the intrinsic multi-disciplinarily complexity (number of distinct knowledge bases), φ_5 is the control complexity (number of closed loops in all control strategies used in the system) and finally φ_6 is the extent of embedded software. A product with increased intelligence has less requirement towards having an extended software. Thus, this sub-criterion depends on the degree of the system intelligence such that

$$\varphi_6 = 1 - \overline{\mathrm{MIQ}}. \tag{11}$$

These types of complexity sources mostly come from interactions among design parameters and physical phenomena, which sometimes unexpectedly cause undesired matters that can result in design or system failures. After determination and normalization of $\mathbf{\Phi}$, and using a linear summation of weighted factors, the complexity value of the concept will be assessed as follows:

$$\mathrm{CX} = \sum_{j=1}^{6} w_j \bar{\varphi}_j, \tag{12}$$

where w_j are the weights associated to each complexity components to of reflect the designer's preferences. It is worth mentioning that fuzzy measures can be incorporated to reflect the codependence among sub-criteria as well.

2.4 Design flexibility

Despite the traditional design approaches, which define a task and create a design for a specific situation, considering flexibility in design leads to producing adaptable systems. A vector of design flexibility for a mechatronic concept and its members can be presented as follows:

$$\mathbf{\Psi} = \left[\psi_1,\ \psi_2,\ \psi_3,\ \psi_4,\ \psi_5\right]^T, \tag{13}$$

in which ψ_1 is the component design flexibility (the number of alternative component design paths), ψ_2 is the customization flexibility (the number of customization options for components), ψ_3 is the number of choices for system architecture, ψ_4 is the interface flexibility (the number of

choices for user interface) and ψ_5 is the control flexibility (the number of options for control algorithms). After determination and normalization of $\mathbf{\Psi}$, the flexibility score for the corresponding concept can be calculated as

$$\mathrm{FX} = \sum_{j=1}^{3} \rho_j \bar{\psi}_j, \tag{14}$$

where ρ_j are the product-dependent weighting factors defined by the designer.

Using the proposed metrics above, we can rapidly quantify the quality of the conceptual design in terms of design flexibility. However, we acknowledge the fact that this process can be done more analytically. For more information, readers are encouraged to study the work of Engel et al. on design adaptability in (Engel et al. 2017) and (Engel and Reich 2015).

2.5 Cost

To achieve an efficient project cost control and management, an estimation of the total cost of a system in the early stages of design is necessary. Managing and reducing the cost of a system at the conceptual design stage is more effective than at the manufacturing and development stage. According to a number of authors (Corbett and Crookall 1986; Mileham et al. 1993), 70–80% of a product cost is determined during the early design stage.

There are various methods for evaluating design and manufacturing costs of a system such as parametric, neural networks and feature-based methods. Among all, using artificial neural network (ANN) seems to be particularly effective for cost estimation of complex systems in which the relationships between cost and design variables cannot be expressed by simple mathematical expressions (Roy 2003; Hegazy and Ayed 1998; Kim et al. 2005) (Fazlollahtabar and Mahdavi-Amiri 2010; Kaushik et al. 1969). Moreover, data-driven approaches can be also used to estimate the cost of a concept. Takai (2009) have proposed a case-based reasoning (CBR) approach which consists of storing information about various products in a knowledge-based and then retrieving a cluster of products that are similar to the concept, and finally adapting the cost of the retrieved products to construct a distribution of the cost of concept.

In this paper, the term "cost" (CT) in mechatronic multi-criteria profile (MMP) will be simply calculated as

$$\mathrm{CT} = \sum_{i=1}^{4} \mathrm{CT}_i, \tag{15}$$

in which CT_1 is the component costs as the summation of prices of all subsystems, CT_2 is the development and manufacturing cost, CT_3 is the integration cost and CT_4 is the failure, repair and overhaul cost.

3 Multi-criteria decision support and aggregation of criteria

After specifying quantitative values for all MMP criteria and corresponding subsets, an effective evaluation technique is needed for choosing among all possible conceptual choices. Thus, a global concept score (GCS) as a multi-criteria design evaluation index can be defined as follows to enable the designers to compare between the feasible generated design concepts;

$$\mathrm{GCS} = S\left(m_1^*, m_2^*, \ \ldots, \ m_n^*\right), \tag{16}$$

where $S(.)$ represents an aggregation function, m_i is the value of the ith criterion from MMP and m_i^* is the corresponding normalized score. For a criterion where a larger value is preferred (e.g., MIQ, flexibility and reliability), we have

$$m_i^* = \frac{m_i}{\max\left(m_i\right)}, \tag{17}$$

and for a criterion that the smaller value is more desirable (e.g., cost and complexity), we can write

$$m_j^* = \frac{1}{\left(\frac{m_j}{\min(m_j)}\right)}. \tag{18}$$

The max() and min() operators from above are evaluating the exact same criterion for different alternatives. For instance, for the MIQ of an alternative, the normalize value is the value of MIQ for that design alternative over the maximum MIQ value among all alternatives.

There are a variety of research efforts in providing aggregation operators for decision analysis based on fuzzy set theory in the presence of dependant and interacting criteria (Grabisch 1996; Marichal and Roubens 1998). In this work, we consider and compare three different aggregation operators based on well-known Choquet and Sugeno fuzzy integrals and also a fuzzy-based multi-layer neural network to calculate the global concept score for each design concept.

3.1 Choquet integrals

As the first aggregation technique, a nonlinear Choquet integral (Marichal 2000, 2002) can be used to compute the global concept score for each feasible generated design concepts. Choquet integral provides a weighting factor for each criterion, and also for each subset of criteria. Using Choquet integrals is a very effective way to measure an expected utility when dealing with uncertainty (Narukawa and Murofushi 2004), which is the case in design in general and mechatronic design in particular. Using this technique and by defining a weighting factor for each subset of criteria, the interactions between

Table 1 Fuzzy interactions and measurements

#	Description of interaction	Fuzzy measurement
I	Negative correlation	$\mu(i,j) > \mu(i) + \mu(j)$
II	Positive correlation	$\mu(i,j) < \mu(i) + \mu(j)$
III	Substitution	$\underset{T\subseteq Y\setminus i,j}{\mu(T)} < \left\{\begin{matrix}\mu(T\cup i)\\ \mu(T\cup j)\end{matrix}\right\} = \mu(T\cup i\cup j)$
IV	Veto effect	$\mu(T)\approx 0$ *if* $T\subset Y, i\notin T$
V	Pass effect	$\mu(T)\approx 1$ if $T\subset Y, i\in T$
VI	Complementarity	$\underset{T\subseteq Y\setminus i,j}{\mu(T)} < \left\{\begin{matrix}\mu(T\cup i)\\ \mu(T\cup j)\end{matrix}\right\} < \mu(T\cup i\cup j)$

multiple objectives and criteria can be easily taken into account along with the designer's intuition. The weighting factor of a subset of criteria is represented by a fuzzy measure on the universe N satisfying the following fuzzy measures (μ) equations:

$$\mu(\phi) = 0, \quad \mu(N) = 1, \tag{19}$$

$$A \subseteq B \subseteq N \ \rightarrow \mu(A) \leq \mu(B), \tag{20}$$

where A and B represent the fuzzy sets. Table 1 shows the most common semantic interactions and the corresponding fuzzy measures.

Thus, GCS can be re-written as follows:

$$\begin{aligned} GCS &= C_\mu\left(m_1^*, m_2^*, \ \ldots, \ m_n^*\right) \\ &= \left(\sum_{i=1}^{n} \phi(\mu, \ i) m_i^* - \frac{1}{2} \sum_{\{i,\, j\}\subseteq N} I(\mu, ij)\left|m_i^* - m_j^*\right|\right), \end{aligned} \tag{21}$$

in which $\phi(\mu, \ i)$ is the importance of criterion i and computed by the Shapley value (ϕ) (Marichal 2002), which is defined as

$$\phi(\mu, i) = \sum_{T\subseteq N\setminus i} \frac{(n-t-1)!t!}{n!}[\mu(T\cup i) - \mu(T)]. \tag{22}$$

where T is a subset of criteria. Furthermore, $\phi(\mu, \ ij)$ is the interaction index between criteria i and j and is defined as follows (Marichal 2002):

$$\begin{aligned} I(\mu, ij) = &\sum_{T\subseteq N\setminus i,j} \frac{(n-t-2)!t!}{(n-1)!} \\ &\times \left[\mu(T\cup ij) - \mu(T\cup i) - \mu(T\cup j) + \mu(T)\right]. \end{aligned} \tag{23}$$

A Choquet fuzzy integral behaves like parameterized aggregator; it can cover the range between min and max of the input level by assigning the density values (fuzzy measures) appropriately. It is important to note that the resulting importance $\phi(\mu, i)$ and interaction indices $I(\mu, ij)$ should satisfy the designer's preferences.

3.2 Sugeno fuzzy

Using a Sugeno fuzzy integral as a method of aggregation of interaction criteria can yield to a relative global concept score (GCS) as follows:

$$\text{GCS} = S_\mu\left(m_1^*, m_2^*, \ \ldots, \ m_n^*\right) = \bigcup_{i=1}^{n}\left[m_i^* \wedge \mu\left(A_{(i)}\right)\right], \tag{24}$$

where (.) in $A_{(i)}$ indicates a permutation on N such that $m_1^* \leq m_2^* \leq \ldots \leq m_n^*$. Also, $\wedge$:= min and $\vee$:= max. Furthermore, $A_{(i)} = \{(i), \ \ldots, \ (n)\}$ and $A_{(n+1)} = \emptyset$. For example, for a decision with five criteria, we have $A_{(3)} = \{3, \ 4, \ 5\}$ and $\mu_{(3)} = \mu_{345}$.

3.3 Neural networks

As the third technique, a Choquet integral-based neural network is presented to be used for the aggregation of criteria and fitting the intuitive requirements for decision-making in the presence of interacting criteria. These networks are highly flexible and their parameters are learned through training (Krishnapuram and Lee 1992; Golmohammadi 2011). The structure of a Choquet integral-based neural node is illustrated in Fig. 3 in which the output is a Choquet integral value (Jung-Hsien 1999).

In this new structure, each neuron is an operator of Choquet integral of $m: N \rightarrow \mathbb{R}$ with respect to μ such that

$$Y_k = C_\mu^{NN}\left(m_1^*, m_2^*, \ \ldots, \ m_n^*\right) = \sum_{i=1}^{n} m_i\left[\mu\left(A_{(i)}\right) - \mu\left(A_{(i+1)}\right)\right], \tag{25}$$

where (.) indicates a permutation on N such that $m_1 \leq m_2 \leq \cdots \leq m_n$. Also, $A_{(i)} = \{(i), \ \ldots, \ (n)\}$ and $A_{(n+1)} = \phi$.

In addition to the fuzzy measure conditions described in Eqs. (19–20), the fuzzy measures used in Eq. (25) satisfy the following additional properties where

$$A \cap B = \phi, \tag{26}$$

$$\mu(A\cup B) = \mu(A) + \mu(B) + \xi\ \mu(A)\mu(B), \tag{27}$$

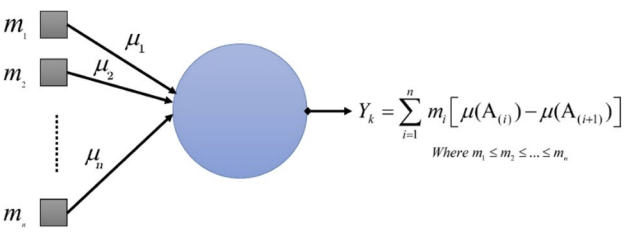


Fig. 3 A fuzzy integral-based neural node where inputs are information sources and output is a fuzzy integral value

in which ξ is the corrective coefficient where $\xi > -1$ and satisfies the following equation:

$$\xi + 1 = \prod_{i=1}^{n} \left(1 + \xi \mu_i\right). \tag{28}$$

The value of $\mu\left(A_{(i)}\right)$ can be determined by a recursive solution of the following equations which use Eq. (27):

$$\mu\left(A_{(n)}\right) = \mu_n, \tag{29}$$

$$\mu\left(A_{(i)}\right) = \mu_i + \mu\left(A_{(i+1)}\right) + \xi \mu_i \mu\left(A_{(i+1)}\right) \quad \text{for } 1 \le i \le n. \tag{30}$$

The proposed neural network is called a fully connected network such that the input layer projects onto the hidden layers of Choquet integral-based neurons and the output of hidden layers projects onto the output layer (Jian and Song 2004). As illustrated before in Fig. 3, in this network, each neuron is represented by a set of linear synaptic links and a fuzzy integral function with respect to a certain fuzzy measure. The synaptic links of a neuron can be interpreted as the degree of importance of the respective input signals. In other words, in this network, a typical "directed link" originates at node i and ends at node k. It has an associated transfer function of Choquet fuzzy integral that specifies the way the output value Y_k at node k depends on the input signal and the "synaptic" links. We assume that the training data for an output node consists of n sets of inputs:$m_1^k,\ m_2^k,\ \ldots, m_m^k$ and with q corresponding desired outputs: $Y_1^k,\ Y_2^k,\ \ldots, Y_m^k$. These desired values can be fed into the network based on a previously studied design case or a case where the aggregation was performed using another tool. The training process is achieved through the well-known backward-error propagation algorithm (Hecht-Nielsen 1989; Freeman and Skapura 1991). With regards to a neural network, a supervised training is a process in which a desired output value for each input pattern is presented. In other words, in supervised training, both the inputs and the outputs are provided. The network then processes the inputs and compares its resulting outputs against the desired outputs. Errors are then propagated back through the system, causing the system to adjust the measures, which control the network. Finally, the best set of fuzzy measures is determined for a node in such a way that the difference between the desired and actual fuzzy integral behavior is minimized. The error is represented as

$$E = \sum_k E_k = \sum_k \sum_i \left(C_i^k - Y_i^k\right)^2, \tag{31}$$

where C_i^k is the Choquet fuzzy integral function of the kth training input vector $\left(m_1^k,\ m_2^k,\ \ldots, m_m^k\right)$ with respect to fuzzy measures μ. Using Eq. (25), we can write C_i^k as follows:

$$C_i^k = \sum_{j=1}^{n} m_j^k \left[\mu\left(A_{(i)}\right) - \mu\left(A_{(i+1)}\right)\right]. \tag{32}$$

Using Eq. (29), we can obtain

$$C_i^k = \sum_{j=1}^{n} m_j^k\ \mu_{ij}\left(1 + \xi^k \mu\left(A_{(i+1)}\right)\right),\quad i = 1, 2, \ldots, q, \tag{33}$$

where n is the number of the input nodes which here is the number of criteria and sub-criteria as illustrated in Fig. 4. Moreover, q is the number of output and we also have $\left(m_1^k \le m_2^k \le \cdots \le m_m^k\right)$. μ_{ij} is the synaptic weight connecting input node j to fuzzy integral i. The network is then optimized by minimizing with respect to the synaptic weights (fuzzy measures) of the network. Thus, we update the fuzzy measures, μ_{ij} using the following equations based on gradient descent:

$$\mu_{ij}^* = \mu_{ij} - \gamma . \frac{\partial E}{\partial \mu_{ij}}, \tag{34}$$

where μ_{ij}^* is the new value for μ_{ij} and for $\partial E / \partial \mu_{ij}$, we have

$$\frac{\partial E}{\partial \mu_{ij}} = 2 \sum_{k=1}^{q} \left(C_i^k - Y_i^k\right) \frac{\partial C_i^k}{\partial \mu_{ij}}, \tag{35}$$

where $\partial C_i^k / \partial \mu_{ij}$ can be obtained from

$$\frac{\partial C_i^k}{\partial \mu_{ij}} = \frac{\partial}{\partial \mu_{ij}} \left(\sum_{j=1}^{n} m_j^k\ \mu_{ij}\left(1 + \xi^k \mu\left(A_{(i+1)}\right)\right) \right). \tag{36}$$

The choice of γ, as a suitable positive constant, is important and it determines the reliability and also time of convergence (Hunt et al. 1992). This parameter can be determined recursively to reach a better convergence time. The learning process is repeated until the changes in μ_{ij} are inferior to a specified small number. The training procedure can be extended to a multi-layer network where there are several nodes arranged in a hierarchical manner. In this case, the training data normally consists of input values at the first layer and the desired output at the last layer. Accordingly, it is possible to compute the fuzzy integral functions and their parameters. After defining the base of the decision-making neural network, now we can intuitively build the structure of the fuzzy-based neural network for newly introduced MMP. In previous aggregation methods, i.e., Choquet and Sugeno, we used weighted sum in sub-criteria aggregation then on the main five criteria we performed a fuzzy aggregation. In the proposed neural network structure, we have avoided weighted sum and we performed a fuzzy aggregation at the same level

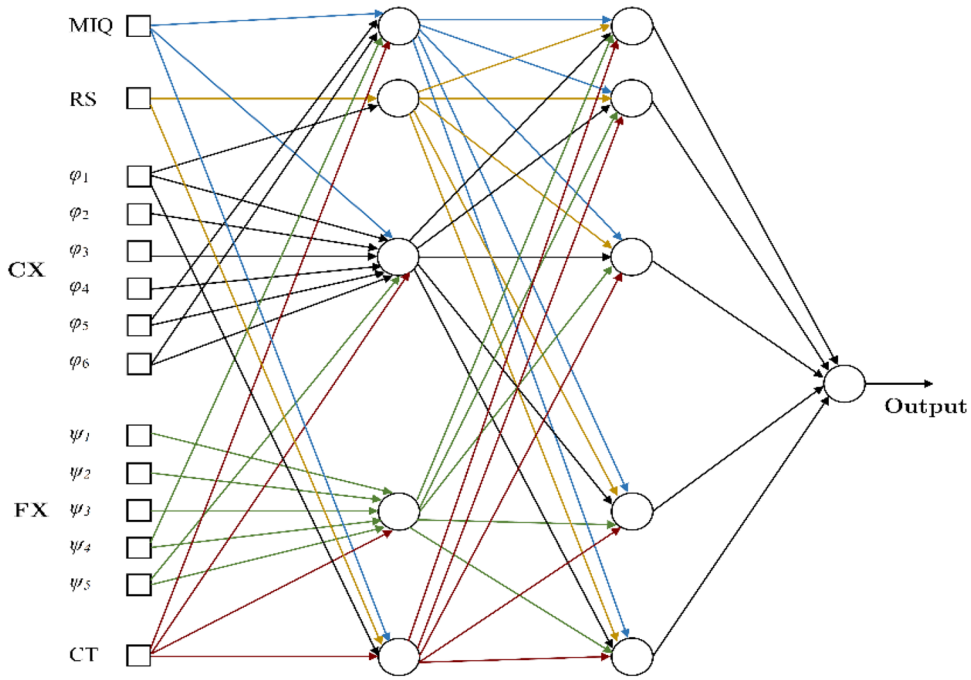


Fig. 4 The structure of the fuzzy integral-based MMP neural network with two hidden layer

for criteria and sub-criteria. 56 densities should be identified through the training process and the initial values fed into the network are intuitively chosen. The corresponding global concept score can be calculated as follows:

$$\mathrm{GCS} = C_{\mu}^{NN}\left(\mathrm{MMP}^{*}\right), \tag{37}$$

where MMP^{*} indicates that all input criteria and sub-criteria to the neural network contain normalized values and are between 0 and 1. After assessing global concept scores (GCS) for each design alternative based on three suggested approaches, the concept selection can be performed to choose the best available design concept and consequently, in a recursive manner, modifications can be applied to the selected concept and subsystems. Based on the requirements and objectives defined in the first stage, a more precise concept selection can be performed and after a few iterations the final decision for the conceptual design can be made. The procedure of conceptual design using the proposed methodology and based on the calculated global concept score is described in flowchart shown in Fig. 5.

4 Case study: design of a vision-guided quadrotor drone

Conceptual design is a crucial stage in a product design process. To in find the best solution among a variety of possible configurations, it is less practical and efficient to search a complex design space in just one iteration. In conceptual design and based on different knowledge bases involved, a complex design space is divided into several subspaces, which also correspond to conceptual alternatives. By performing a proper evaluation for all of these subspaces, the designer then is able to narrow down the search space to one or two of the subspaces.

Autonomous systems and robotic machineries have been increasingly employed in various industrial, urban and exploratory applications during the last decades. However, these types of systems are generally limited to operate in highly structured environments. Thus, integration of vision sensors with automatic systems and generally "visual servoing" systems helped solve this problem by producing non-contact and wide measurements of the working area for the machine (Corke 1993). As a case study of using mechatronic multi-criteria profile (MMP) for conceptual design of a mechatronic system, design of a Quadrotor drone along with a visual servoing system for tracking and following a moving target (Mohebbi et al. 2014a, b, c) is presented here. As a mechatronic system, design of a Quadrotor drone can be considered as a non-trivial problem which involves a medium-level complexity and allows several alternatives in both structure and controller design. The objective here is to design a quadrotor with a visual servoing system which is capable of tracking and following a moving object (x_k) with the maximum velocity of 1 m/s, starting within 3 s after the object enters the vision system's field of sight and also within the area of motion with dimensions of 3000 mm × 3000 mm × 1000 mm.

Based on the defined objective(s), the concept selection and design should be performed upon deciding for the mechanical structure, actuators, position and velocity

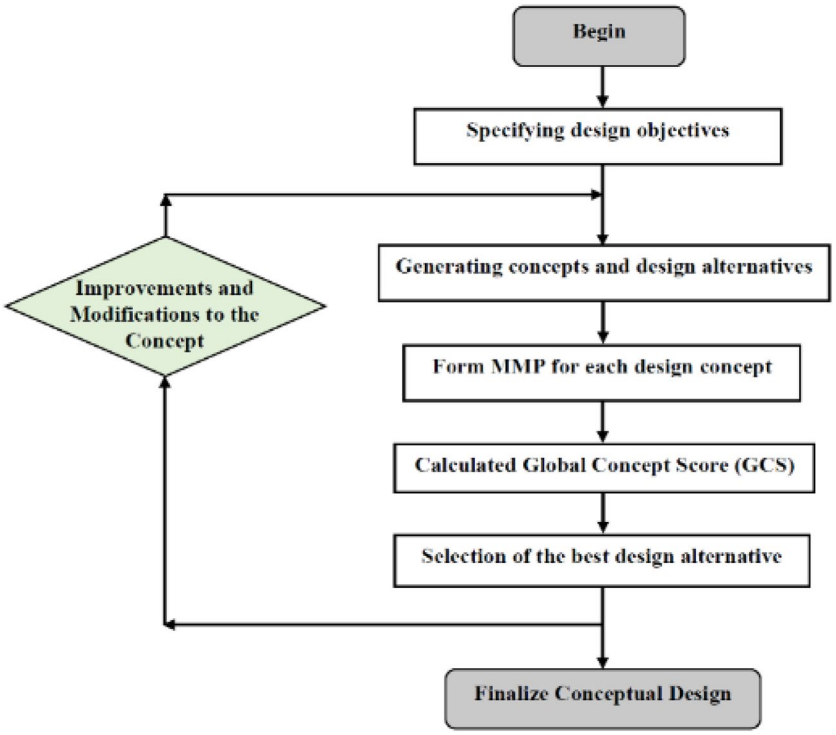


Fig. 5 Conceptual design of a mechatronic system based on MMP and global concept score (GCS)

sensors, vision sensors, controllers, and battery as the most essential components of the system. After a careful identification of feasible subsystems and components by the designer, the generation of the design alternatives can be performed. The main responsibility of the decision-making task in conceptual design is to select between all possible alternatives and subsystems which better match the design objectives. The generation of design concepts and alternatives for a Quadrotor system is described in Fig. 6. Based on the number of options for each subsystem (material, structure, actuator, battery, vision and position sensors, motion control and visual servoing system), we are able to generate 7776 concepts but after a feasibility study and dependency modeling, just 972 concepts were considered as acceptable as some components are not compatible with others. For example, a concept with three cameras should not use a position-based visual servoing controller.

Here and to simplify the case study and based on feasibility in system integration, we have chosen four feasible concepts to study the proposed design method. However, there exist a number of methods to aid the designers with

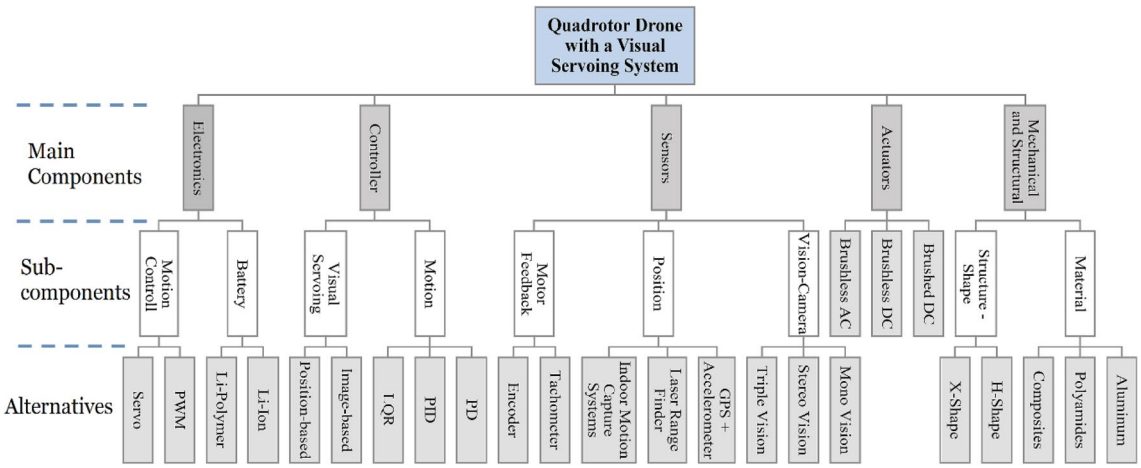


Fig. 6 Parts and subsystems for generation design concepts

generating optimal concepts among the feasible set. For example, Ziv-Av and Reich (2005).

Based on the material used, the frame structure and subsystems selected for one specific concept, the total mass, required power, payload, maximum allowable inertia moment, force and bandwidth can be also estimated (Table 2). An approximation of the total cost can also be identified based on the components and manufacturing process. Table 3 briefly gives the results for the estimated values for proposed concepts. Max power and also max inertia moment were estimated using simplified CAD models for X-shape and H-shape alternatives and by the taking into account the unit weight of the aluminum and polyamides with the maximum vertical speed of 9 m/s. Bandwidth was estimated according to the previously simulated IBVS and PBVS vision-based control systems and cost was estimated using Eq. 15 by taking the approximate cost for the materials, components, visual servoing systems, etc.

Table 3 Estimated design features for generated concepts

	Concept 1	Concept 2	Concept 3	Concept 4
Max power (W)	450	500	350	400
Max inertia moment (kg m^2)	5E-3	5.2E-3	4E-3	4.5E-3
Bandwidth (Hz)	70	70	60	60
Payload (kg)	0.5	0.5	0.6	0.6
Cost (unit)	0.8	1	0.7	0.7

4.1 MMP assessment

4.1.1 Machine intelligence quotient

To measure the machine intelligence quotient (MIQ) based on the proposed method in Sect. 2.1, two ITG models for the quadrotor system with a position-based visual servoing (PBVS) and an image-based visual servoing (IBVS) are considered and depicted in Fig. 7.

To measure the task intelligence costs, a counting agent was built into a computer simulation to measure the controllers' output data units per time (kb/s) and human interface input data per time for each task defined. Thus, for PBVS system, nine different simulations were attempted for each task while counting the input and output transferred data as a measure of required intelligence. Another unit was put in the simulations to count and accumulate data quantities (packets) transferred between subsystems, interface and controllers. The result would form the data transfer matrix, F. Task intelligence cost, data transfer quantity, and interface complexity for both cases are measured, scaled and listed in Table 4. For example, in the PBVS system, $\tau_9 = 12$. This reflects that to perform task T_9, an overall of 120 kb/s was measured for the output and input of the system and scaled to 12. Moreover, we see that $f_{19} = 53$. This means that the data quantity transferred from task 1 to task 9 was measured 53 units.

The MIQ for the case of using a PBVS approach can be calculated as follows through Eqs. (2–8):

$$CIQ = \sum_{i=1}^{n} a_{i1}\tau_i + \sum_{i=1}^{n} a_{i2}\tau_i = \left(\tau_2 + \tau_3 + \tau_4 + \tau_5 + \tau_6 + \tau_9\right) + \left(\tau_1 + \tau_7 + \tau_8\right) = 92, \tag{38}$$

$$\begin{aligned} \mathrm{HIQ} &= \left(\sum_{i=1}^{n} a_{i2}\tau_i\right) + \left(C_{mh}\sum_{i=1}^{n}\sum_{i=1}^{n} a_{i1}a_{j2}f_{ij}\right) + \left(C_{mh}\sum_{i=1}^{n}\sum_{i=1}^{n} a_{i2}a_{j1}f_{ij}\right) \\ &= (\tau_1 + \tau_7 + \tau_8) + C_{mh}(f_{38} + f_{57} + f_{58} + f_{67} + f_{68}) \\ &\quad + C_{mh}(f_{12} + f_{14} + f_{15} + f_{16} + f_{17} + f_{19} + f_{89}) \\ &= 31 + (0.07)181 + (0.08)289 \\ &= 66.79, \end{aligned} \tag{39}$$

$$\mathrm{MIQ}_{pb} = \mathrm{CIQ}_{pb} - \mathrm{HIQ}_{pb} = 92 - 66.79 = 25.21. \tag{40}$$

The same procedure can be used to calculate the MIQ for the case of using an IBVS approach,

Table 2 Design alternatives

	Concept 1	Concept 2	Concept 3	Concept 4
Frame structure	X-shape	H-shape	X-shape	H-shape
Material	Aluminum	Aluminum	Polyamides	Polyamides
Motors	Brushed DC	Brushed DC	Brushless DC	Brushless AC
Motor feedback	Encoder	Tachometer	Encoder	Tachometer
Visual servo	PBVS	PBVS	IBVS	IBVS
Camera config	Mono	Stereo	Stereo	Mono
Motion control	PID	LQR	PID	LQR
Position sensor	GPS + Accel	Motion capture	GPS + Accel	GPS + Accel
Battery	Li-ion	Li-Poly	Li-ion	Li- Poly

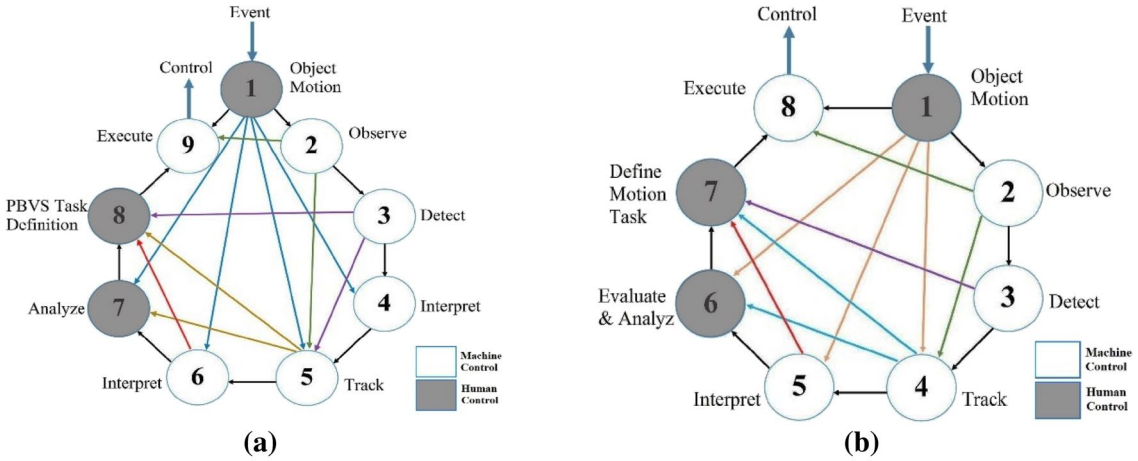


Fig. 7 ITG model for **a** a position-based visual servoing (PBVS) scheme and **b** an image-based visual servoing (IBVS) scheme for a quadrotor system

Table 4 Parameters required for assessing MIQ for two cases of PBVS and IBVS

PBVS	IBVS
$\tau_{pb} = \{\tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7, \tau_8, \tau_9\} = \{4, 6, 11, 10, 12, 10, 17, 10, 12\}$	$\tau_{ib} = \{\tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7, \tau_8\} = \{4, 6, 11, 12, 19, 18, 9, 12\}$
$F_{pb} = \begin{bmatrix} 0 & 77 & 0 & 28 & 42 & 38 & 30 & 0 & 53 \\ 0 & 0 & 61 & 0 & 52 & 0 & 0 & 0 & 39 \\ 0 & 0 & 0 & 59 & 54 & 0 & 0 & 44 & 0 \\ 0 & 0 & 0 & 0 & 51 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 31 & 40 & 39 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 31 & 27 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 37 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 21 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$	$F_{ib} = \begin{bmatrix} 0 & 77 & 0 & 42 & 38 & 32 & 0 & 53 \\ 0 & 0 & 61 & 52 & 0 & 0 & 0 & 37 \\ 0 & 0 & 0 & 59 & 0 & 0 & 47 & 0 \\ 0 & 0 & 0 & 0 & 53 & 39 & 34 & 0 \\ 0 & 0 & 0 & 0 & 0 & 43 & 40 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 40 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 34 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$
$c_{\mathrm{mh}} = 0.07, \quad c_{\mathrm{hm}} = 0.08$	$c_{\mathrm{mh}} = 0.07, \quad c_{\mathrm{hm}} = 0.08.$
$A_{pb} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	$A_{ib} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

$$\mathrm{CIQ} = \sum_{i=1}^{n} a_{i1}\tau_i + \sum_{i=1}^{n} a_{i2}\tau_i = \left(\tau_2 + \tau_3 + \tau_4 + \tau_5 + \tau_6\right) + \left(\tau_1 + \tau_6 + \tau_7\right) = 91, \tag{41}$$

$$\begin{aligned} HIQ &= \left(\sum_{i=1}^{n} a_{i2}\tau_i\right) + \left(C_{mh}\sum_{i=1}^{n}\sum_{i=1}^{n} a_{i1}a_{j2}f_{ij}\right) + \left(C_{mh}\sum_{i=1}^{n}\sum_{i=1}^{n} a_{i2}a_{j1}f_{ij}\right) \\ &= (\tau_1 + \tau_6 + \tau_7) + C_{mh}(f_{46} + f_{56} + f_{37} + f_{47} + f_{57}) \\ &\quad + C_{mh}(f_{12} + f_{14} + f_{15} + f_{16} + f_{17} + f_{78}) = 25 + (0.07)203 + (0.08)276 = 61.29, \end{aligned} \tag{42}$$

Table 5 Mean time to failure (Hrs.) for the components used in the generated concepts

Components	Motors	Motor seedback	Position sensor	Battery
Concept 1	Brushed	Encoder	GPS + Accel	Li-ion
	3810	8230	2920	10,540
Concept 2	Brushed	Tachometer	Motion capture	Li-Poly
	3810	8900	1750	11,780
Concept 3	Brushless	Encoder	GPS + Accel	Li-ion
	5120	8230	2920	10,540
Concept 4	Brushless	Tachometer	GPS + Accel	Li-Poly
	5120	8900	2920	11,780

Table 6 Reliability scores for generated concepts

	Concept 1	Concept 2	Concept 3	Concept 4
Reliability score	$RS_1 = 0.86$	$RS_2 = 0.91$	$RS_3 = 0.93$	$RS_4 = 1$

$$MIQ_{pb} = CIQ_{pb} - HIQ_{pb} = 91 - 61.29 = 29.71. \quad (43)$$

4.1.2 Reliability score

The reliability scores for all the presented design alternatives can be calculated using Eqs. (9–10) and the components listed in Fig. 6 and Table 2. As stated before, in the presented case study and to avoid complexity in assessing MMP for all the design concepts, the dynamic effects for reliability assessment have not been taken into account. The reliability of each concept has been determined based on the failure probability of their components. Although there is not a solid reference to address each component's failure rate, we managed to extract a number of values for sensors and actuators from various resources, i.e., technical manuals, reports and experiment documentations (Jones 2011, Nasar 1981, AB-019: Lifetime of DC vibration motors (MTTF & FIT) 2014) (Table 5). The failure probabilities were considered for a system under 100,000 h of work. Based on information listed in Table 5, the reliability score for each concept can be calculated as shown in Table 6.

4.1.3 Complexity value

Based on the method proposed in Sect. 2.3, complexity values for all the design concepts can be determined. As an example, we discuss the assessment of the complexity score for concept 1 here. The first sub-criteria, φ_1, is the number of components and can be obtained by listing all the necessary sub-components based on the choice of the design alternatives and their subsystems. For the first concept, $\varphi_1^1 = 24$ as we have 4 motors, 4 gearboxes, 4 propellers, 1 GPS, 1 Accelerometer, 1 gyroscope, 1 Radio transmitter and receiver, 4 encoders, 1 battery, 1 flight control and driver system, 1 frame and 1 camera. φ_2, as the degree of architecture complexity is obtained from the number of design interconnections. Figure 8 illustrates these interconnections for concept 1, thus, $\varphi_2^1 = 12$.

φ_3 is the design process complexity which is specified by the number of backtracks in the design process. Figure 9 shows these design backtracks for concept 1. Accordingly, we achieve $\varphi_3^1 = 5$.

Moreover, φ_4 is the intrinsic multi-disciplinarily complexity which is specified by the number of distinct knowledge bases. For the first concept, we have $\varphi_4^1 = 5$, as we are dealing with aerodynamics, mechanical design, electronics, flight control and visual servoing knowledge bases. φ_5 is the

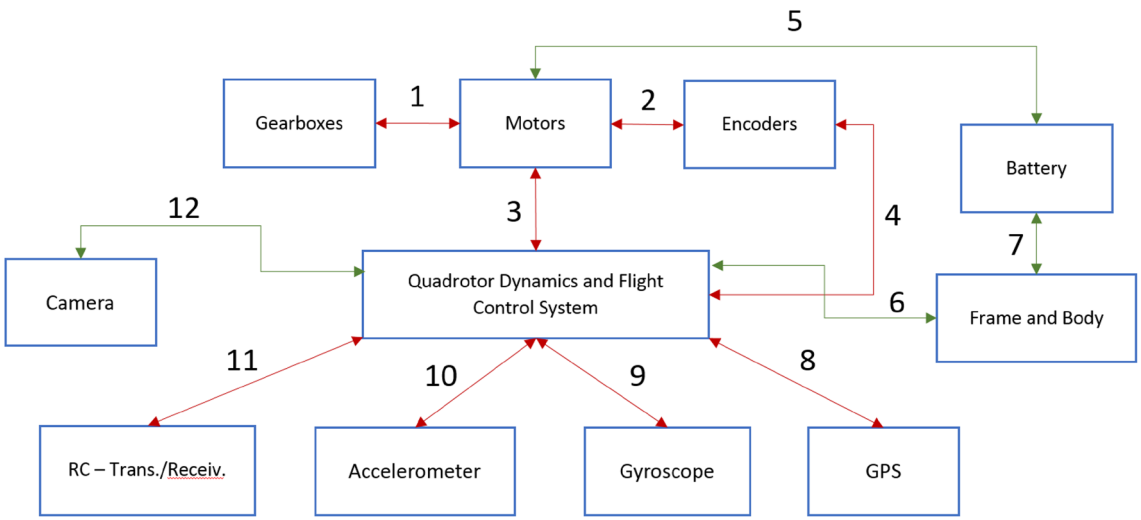


Fig. 8 Design interconnections for concept 1

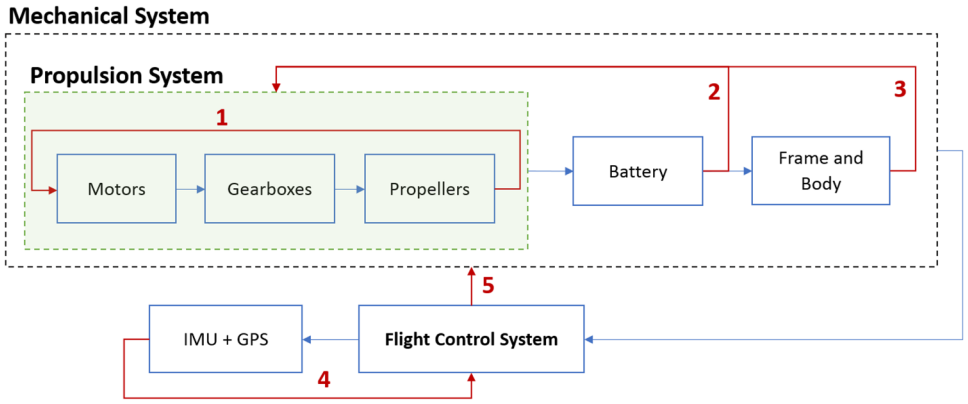


Fig. 9 Design process backtracks for concept 1

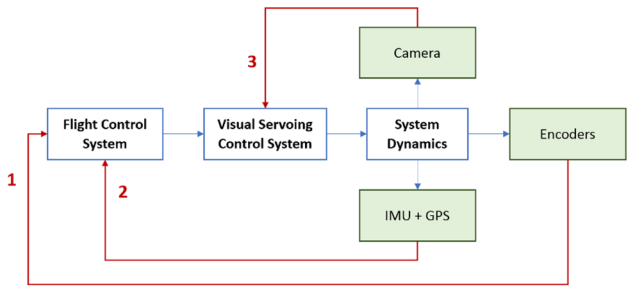


Fig. 10 Closed-loop (feedback) control systems in concept 1

control complexity as is measures as the number of closed loops in all control strategies used in the system. Figure 10 specifies these closed-loop systems is concept 1 so that we get $\varphi_4^1 = 3$.

Finally, φ_6 is the extent of embedded software which depends on the degree of the system intelligence such that $\varphi_6 = 1 - \overline{\text{MIQ}} = 1 - (25.21/29.71) \simeq 0.16$.

The values for each members of complexity vector and the final complexity score are shown in Table 7, where Φ_i are the complexity scores and $\bar{\Phi}_i$ are the normalized values for which $0 \leq \bar{\Phi}_i \leq 1$ and for a concept with minimum complexity, $\bar{\Phi}_i = 1$. Furthermore, w_j are the weights associated to the complexity components, and considered to be equal to 0.17 for $i = 1, \ldots, 5$, and 0.15 for $i = 6$. Based on Eq. (18), for an alternative with $\varphi_6 = 0$, we have $\overline{\varphi_6} = \frac{1}{0/0} \rightarrow 1$ and for an alternative with $\varphi_6 = 0.16$, we have $\overline{\varphi_6} = \frac{1}{0.16/0} \rightarrow \frac{1}{\infty} \rightarrow 0$.

As mentioned in Sect. 3, for the complexity criterion where the smaller value is more desirable for a conceptual design, the values for sub-criteria are normalized in a way that a concept with less complexity gains better complexity score. From Table 7, we can observe that concept 2 has lower complexity score compared to the others as it has more components and subsystems, more interconnections in design architecture, more loops in design process, more knowledge bases involved and finally higher control system complexity.

4.1.4 Flexibility value

The final results for flexibility assessment based on the proposed method in Sect. 2.4 are presented in Table 7, where Ψ_i are the flexibility scores and $\bar{\Psi}_i$ are the normalized values which $0 \leq \bar{\Psi}_i \leq 1$. $\rho_i = [0.3, \quad 0.2, \quad 0.1, \quad 0.1, \quad 0.3]$ are the weights associated to the flexibility components to assess the final flexibility score. Ψ_i are calculated based on the number of alternative component designs, the number of customization options for components, the number of choices for system architecture, user interface and also for control algorithms. For example, the flexibility values for the alternative concept 1 are considered as follows:

For ψ_1 as the component design flexibility, we have two options of Time-of-Flight or RGB for the monocular camera, three options of using feature points, image moments and lines for image features tracked in PBVS visual servoing control, and for encoders, we have two alternatives of magnetic rotary or optical rotary encoders which makes the design to have $3 \times 2 \times 2 = 12$ different paths. For

Table 7 Complexity values for design alternatives

Concept 1	Concept 2	Concept 3	Concept 4
$\Phi_1 = [24 \quad 12 \quad 5 \quad 5 \quad 3 \quad 0.16]^T$	$\Phi_2 = [32 \quad 16 \quad 5 \quad 6 \quad 4 \quad 0.16]^T$	$\Phi_3 = [25 \quad 13 \quad 5 \quad 5 \quad 4 \quad 0]^T$	$\Phi_4 = [25 \quad 14 \quad 5 \quad 6 \quad 3 \quad 0]^T$
$\bar{\Phi}_1 = [1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0]^T$	$\bar{\Phi}_2 = [0.75 \quad 0.75 \quad 0.83 \quad 0.75 \quad 0]^T$	$\bar{\Phi}_3 = [0.96 \quad 0.92 \quad 1 \quad 1 \quad 0.75 \quad 1]^T$	$\bar{\Phi}_4 = [0.96 \quad 0.86 \quad 1 \quad 0.83 \quad 0.75 \quad 1]^T$
$\text{CX}_1 = 0.850$	$\text{CX}_2 = 0.693$	$\text{CX}_3 = 0.937$	$\text{CX}_4 = 0.898$

Table 8 Flexibility values for design alternatives

Concept 1	Concept 2	Concept 3	Concept 4
$\Psi_1 = [12\ \ 18\ \ 3\ \ 4\ \ 2]^T$	$\Psi_2 = [17\ \ 18\ \ 2\ \ 4\ \ 2]^T$	$\Psi_3 = [12\ \ 18\ \ 3\ \ 4\ \ 2]^T$	$\Psi_4 = [14\ \ 15\ \ 2\ \ 4\ \ 2]^T$
$\bar{\Psi}_1 = [0.705\ \ 1\ \ 1\ \ 1\ \ 1]^T$	$\bar{\Psi}_2 = [1\ \ 1\ \ 0.667\ \ 1\ \ 1]^T$	$\bar{\Psi}_3 = [0.705\ \ 1\ \ 1\ \ 1\ \ 1]^T$	$\bar{\Psi}_4 = [0.823\ \ 0.833\ \ 0.667\ \ 1\ \ 1]^T$
$FX_1 = 0.911$	$FX_2 = 0.967$	$FX_3 = 0.911$	$FX_4 = 0.880$

Table 9 Total cost factors for generated concepts

Concept 1	Concept 2	Concept 3	Concept 4
$CT_1 = 1$	$CT_2 = 0.78$	$CT_3 = 0.94$	$CT_4 = 0.91$

ψ_2 as the customization flexibility, we can place the rotors in upward or downward position or with upward flapping. Moreover, the weight of the components, i.e., batteries, electrical boards, etc., can be distributed on the whole frame or can be concentrated on the center of the frame hull. Finally, for the frames with the same weight we have three options of high, medium or low inertia. This makes the customization options to be equal to $3 \times 2 \times 3 = 18$. For the system architecture, ψ_3 is the number of choices, and for concept 1, we have three options of design for visual tracking, design for flight durability, and design for flight control. Furthermore, ψ_4 is the number of choices for user interface which for concept 1, we have four options of joystick, mobile platform, VR-headset, or a pc-enabled station. Lastly, for the control flexibility, ψ_5, we have two options of PD and PI/PID controller for the PID flight control system.

From the results in Table 8, we can assume that the concepts represent almost the same flexibility level according to the proposed assessment method.

4.1.5 Overall cost

According to Eq. 16 and considering a set of standard and commonly used components, the cost factor for each concept can be calculated as shown in Table 8. For a concept with "minimum total cost", $CT = 1$ (Table 9).

4.2 Aggregation of criteria and global concept scores

By taking into account the five design criteria as elements of MMP, one can conclude that $2^5 = 32$ fuzzy measures should be specified to be used in Choquet and Sugeno aggregation functions. Equation (19) identifies two of these measures ($\mu_\emptyset = 0,\ \ \mu_{12345} = 1.$) which leaves 30 measures to be specified. In the present case study, the fuzzy measures were obtained in an intuitive manner by the authors and a group of 30 researchers (all specialized in system design and mechatronics) through a questionnaire (See "Appendix"). This questionnaire collects the following information from various design point of views;

Table 10 Criteria interactions and the correlation measure

Interactions	Correlation measure
Positive correlation	$\gamma_{ij} > 0$
Negative correlation	$\gamma_{ij} < 0$
No correlation	$\gamma_{ij} = 0$

Table 11 30 fuzzy measures for the case study of conceptual design of a Quadrotor drone with a visual servoing system

$\mu_1 = 0.23$	$\mu_{12} = 0.45$	$\mu_{13} = 0.47$	$\mu_{14} = 0.34$	$\mu_{15} = 0.51$
$\mu_{123} = 0.61$	$\mu_2 = 0.29$	$\mu_{23} = 0.52$	$\mu_{24} = 0.42$	$\mu_{25} = 0.56$
$\mu_{124} = 0.60$	$\mu_{135} = 0.69$	$\mu_3 = 0.17$	$\mu_{34} = 0.35$	$\mu_{35} = 0.33$
$\mu_{125} = 0.67$	$\mu_{145} = 0.67$	$\mu_{245} = 0.73$	$\mu_4 = 0.16$	$\mu_{45} = 0.41$
$\mu_{134} = 0.63$	$\mu_{234} = 0.68$	$\mu_{345} = 0.49$	$\mu_{235} = 0.62$	$\mu_5 = 0.22$
$\mu_{1234} = 0.77$	$\mu_{1235} = 0.84$	$\mu_{1345} = 0.84$	$\mu_{2345} = 0.78$	$\mu_{1245} = 0.82$

The degree of importance for each of the mentioned five criteria in designing a good mechatronic product (i.e., μ_i , $1 \le i \le 5$).

The degree of correlation between each pair of criteria or the effect of increasing criterion i on criterion j (Table 10), i.e., γ_{ij} , $1 \le i, j \le 5$, which specifies the fuzzy measures as follows:

$$\mu_{ij} = \mu_i + \mu_j - \gamma_{ij}, \tag{44}$$

$$\mu_{ijk\ldots} = \mu_i + \mu_j + \mu_k + \ldots - \left(\max\left(\left|\gamma_{ij}\right|,\ 1 \le i,j \le 5\right).\ sign\left(\gamma_{ij}\right)\right). \tag{45}$$

The specified fuzzy measures used in the present case study are shown in Table 11.

Based on what was suggested for the structure of the fuzzy integral-based neural network (Fig. 4), 56 weights have been identified (instead of 2^{14} measures which corresponds to a case which all criteria and sub-criteria are considered in the same level and form a total of 14 criteria) and incorporated for the aggregation of criteria. Table 12 shows the results for the assessed MMP elements and global concept scores related to Choquet, Sugeno, and

Table 12 MMP elements and global concept scores (GCS) based on neural network, Choquet and Sugeno aggregations for all design alternatives

MMP criterion	Concept 1	Concept 2	Concept 3	Concept 4
MIQ	0.84	0.84	1	1
RS	0.86	0.91	0.93	1
CX	0.85	0.69	0.93	0.89
FX	0.91	0.96	0.91	0.88
CT	1	0.78	0.94	0.91
GCS_C	0.89	0.83	0.96	0.94
GCS_S	0.84	0.78	0.91	0.91
GCS_{NN}	0.87	0.82	0.93	0.92

neural network aggregations (GCS_C, GCS_S, GCS_{NN}) for each generated concept and design alternatives for a Quadrotor system.

4.3 Selecting the elite concept and comparison of results

Based on the values indicated in Table 12 for global concept scores (GSC), one can conclude that based on a Choquet integral aggregator and also a neural network-based aggregator, the best selected concept corresponds to the third design alternative in which a stereo vision is used and the visual servoing scheme is an image-based system (IBVS) with a PID controller. In this concept, polyamide is selected as the structure material and an X-shape chassis along with brushless DC motors and encoders is used. The navigation system is based on a GPS and the energy source would be a lithium-ion battery set. On the other hand, the Sugeno aggregator decides that the fourth concept is equally efficient for designing a Quadrotor drone system. In the fourth concept, a monocular vision system is used and the visual servoing scheme is also an image-based system (IBVS) with a LQR controller (linear quadratic regulator). For the structure, an H-shape frame is selected and polyamide is chosen to be the main material for the chassis. Since brushless AC motors are suitable in this concept, an inverter is also needed and the velocity of the shafts is determined using tachometers. For the navigation, the same GPS and accelerometers are used, and the energy source would be a lithium–polymer battery set. From the global concept scores resulted from using neural network aggregation, one can conclude that these two concepts can be almost equally regarded as the right choices to push for detailed analysis and design. However, further study and simulation would better clarify the conclusion. In this case, based on the selected elements and subsystems listed in Table 2, and to validate the results for global concept score for each aggregation function and consequently the design decisions, a computer simulation of a visual servoing system on a Quadrotor drone has been performed. Figure 11 shows the control system architecture used in the simulation process. The models have been built for concept 3 and concept 4 as the best concepts with regards to their global concept scores. The physical parameters of the models were taken into account based on the information from Table 3. Besides the actuator, sensor and battery types, the difference between the selected concepts mostly concerns the camera configuration (stereo vs. mono) and motion controllers (PID vs. LQR). Figures 12, 13 and 14 show the simulation results for a visual servoing system designed based on concepts 3 and 4. The servoing task consists of coinciding four image features to four desired predefined points in the image plane. The initial and desired configurations of the image features for each test are given in Table 13.

As shown in Fig. 12, the system designed based on concept 3 shows a relatively better performance than the one designed based on concept 4. In terms of feature trajectories, concept 3 has less oscillation, overshoot and unnecessary motion than concept 4 (Fig. 12b–d). This means concept 3 is showing better visual tracking performance. This system has also a better tracking convergence time (Fig. 14a, b), as it converges within 13 s and with regards to the motion, the overall frame velocity components for the Quadrotor frame starts at relatively smaller values (Fig. 13a, b) which indicates that the system has less abrupt motion which can have a negative effect on

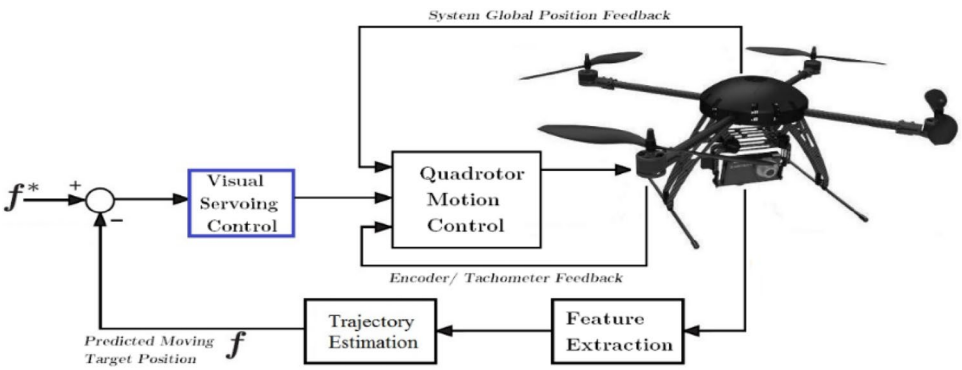


Fig. 11 An integrated visual servo control for a Quadrotor system in a tracking and following use-case

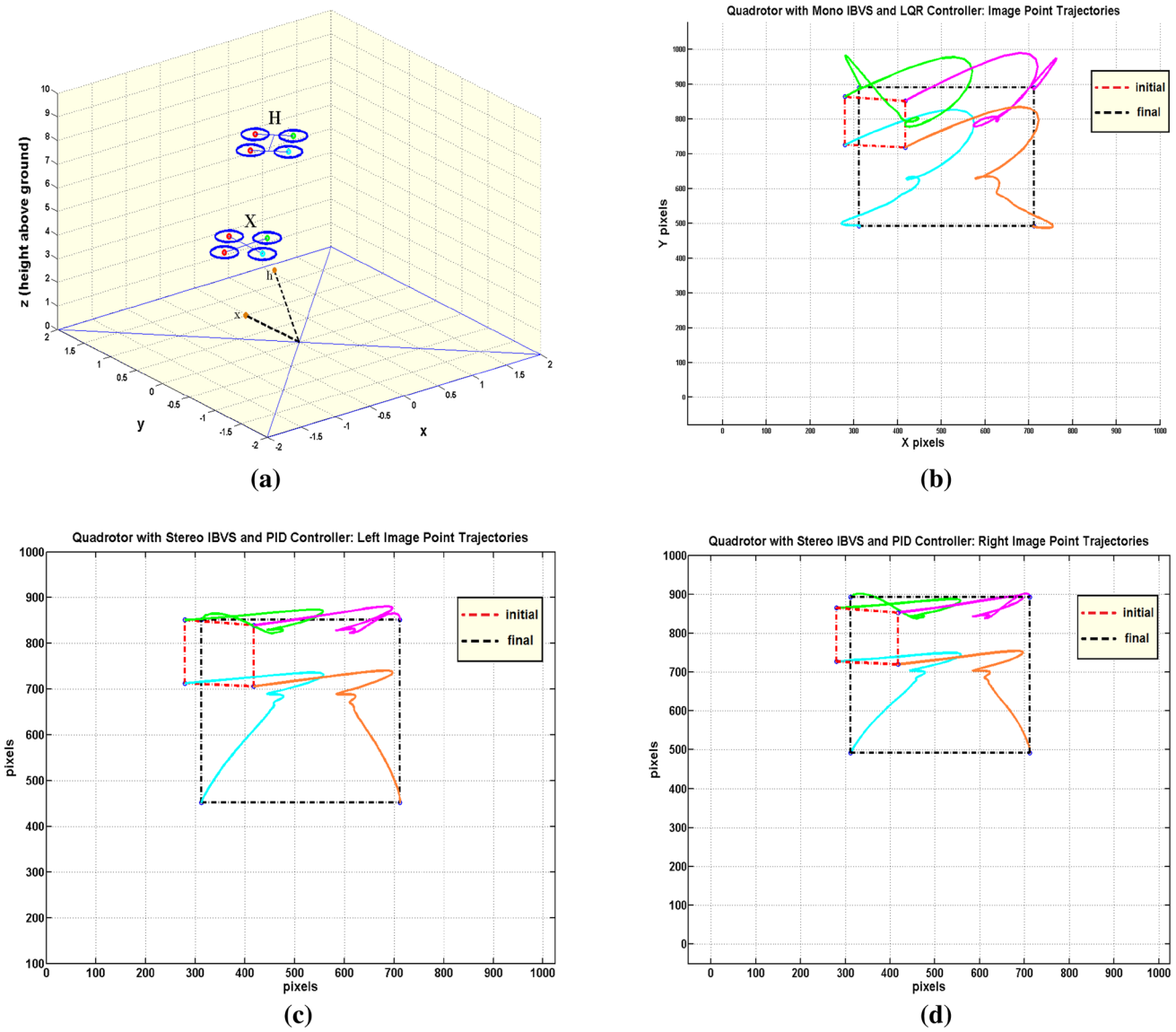


Fig. 12 **a** A simulation model of a Quadrotor drone with a visual servoing system. Image feature trajectories in **b** concept 4 Monocular IBVS with LQR controller, **c, d** concept 3: stereo IBVS with PID controller

the system hardware and total energy consumption. These results indicate that GCS_C and GCS_{NN} have performed a better job than GCS_S in suggesting the best conceptual design candidate.

Choquet integral as a nonlinear fuzzy integral is an effective way for decision-making in the presence of interacting criteria. The overall importance of a criterion and the interaction indices are carefully determined using the Shapely value (Eqs. 24–25). On the other hand, the Sugeno fuzzy integral, more or less performs as a kind of a weighted median. Therefore, it can be inferred that Sugeno integral produces a very conservative value for the global concept score for each design. Although due to the consideration of weight of importance for different sets of criteria, this aggregation function can also be useful for decisions under uncertainty as well as for multi-criteria design problems. Due to the iterative nature and learning process in the neural network-based aggregation method, more reliable and precise values are achieved for the global concept scores. In a more complex design case where more feasible components and subsystems are involved, the elite set of the generated concepts includes larger numbers of design alternatives and the precision of the aggregation function comes in use to choose one final concept. More importantly, in the neural network-based aggregation, not only the interactions between the main criteria are considered, but also the interactions between a specific sub-criterion of a criteria with another sub-criterion of a

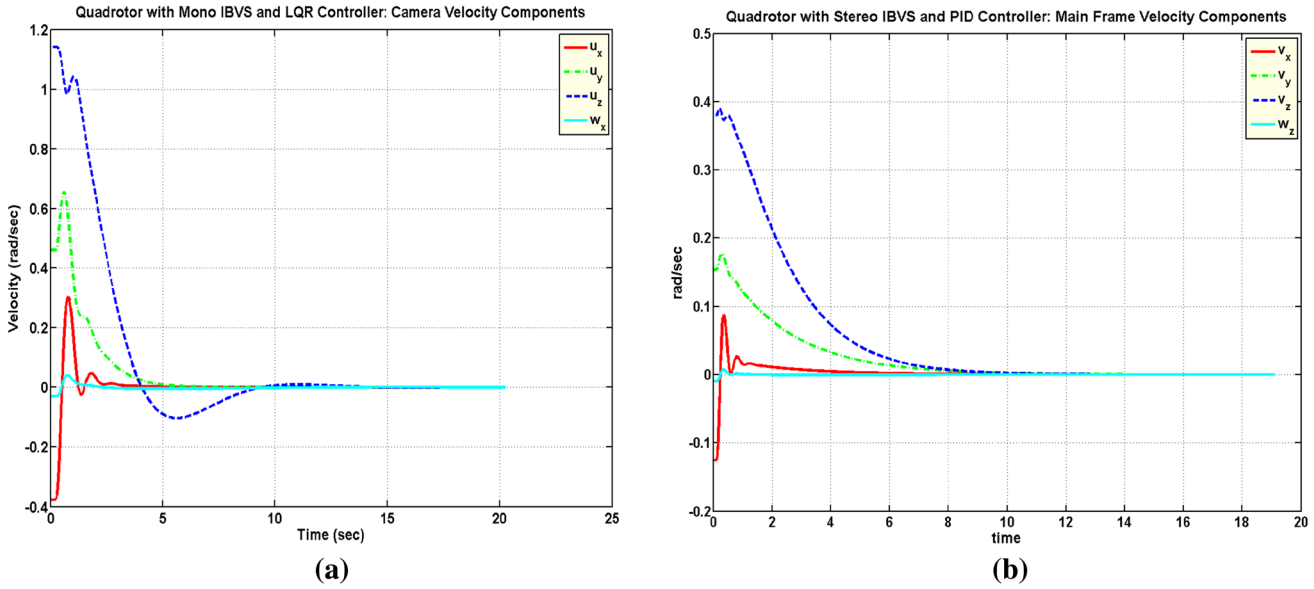


Fig. 13 Camera frame velocity in systems based on concept 4 (**a**) and concept 3 (**b**)

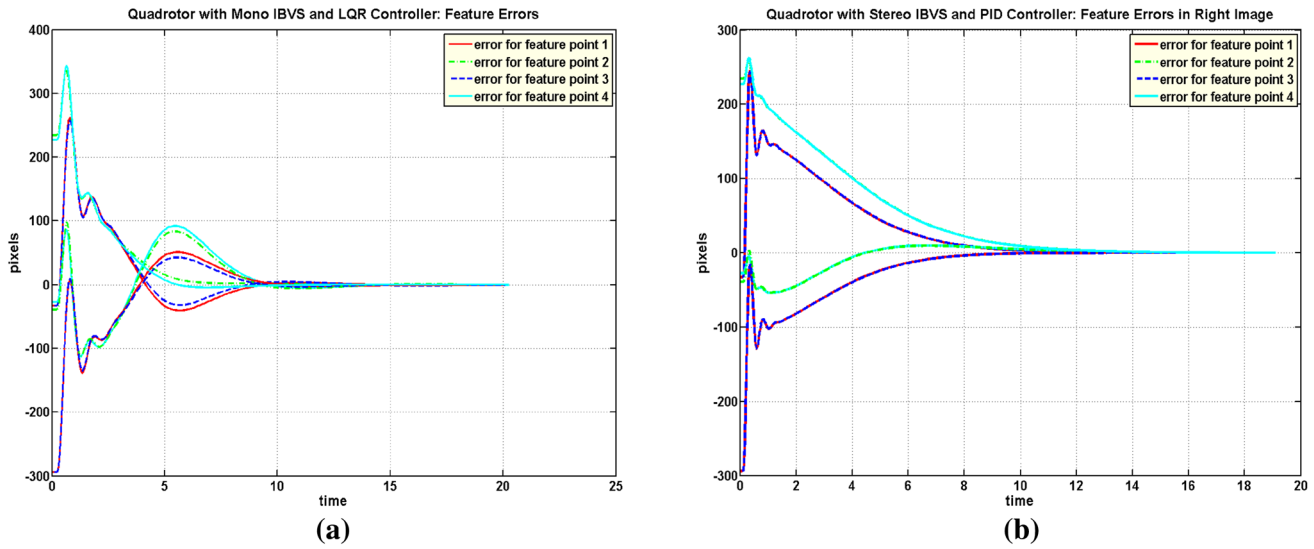


Fig. 14 Image feature errors in systems based on concept 4 (**a**) and concept 3 (**b**)

Table 13 Initial (IR, IL) and desired (DR, DL) location of feature points in pixel for right and left images

Features	Point 1 (x y)		Point 2 (x y)		Point 3 (x y)		Point 4 (x y)	
Initial points								
IR	279	726	279	864	415	853	415	719
IL	281	712	281	851	415	839	415	705
Final points								
DR	312	492	312	892	712	892	712	492
DL	312	452	312	852	712	852	712	452

different criteria is considered and measured. This feature vastly increases the reliability of the aggregation results.

5 Discussion and conclusion

In this paper, a new multi-criteria profile (MMP) has been introduced for concept evaluation in design of mechatronic systems and the assessment procedure has been thoroughly discussed. The MMP consists of five main design criteria such as intelligence, reliability, complexity, flexibility and cost, and can be embedded in an automated design routine. Based on the assessed MMP for each concept and using various aggregation techniques, a global concept score (GCS) has been calculated to ease the procedure of concept evaluation, selection and modification. For the aggregation of criteria, three functions have been introduced based on Choquet integral, Sugeno integral and a fuzzy integral-based neural network. Moreover, the proposed method has been applied to a case study of designing a Quadrotor drone system capable of tracking and following an object using a visual servoing system. At the end, the two selected concepts based on different aggregation functions have been tested and compared using a computer simulation of a visual servoing task. Beside the introduction of MMP and the methods of aggregation, the main contribution of this paper can be stated as providing a systematic approach to support evaluating and finding an optimal conceptual design of a mechatronic system in the presence of large number of interacting criteria. This approach can be used in an automated software platform along with collaboration between designers from various disciplines based on a guideline that the presented methodology has provided.

One of the limitations of the proposed approach is that there is a large number of fuzzy measures and also parameters used in criteria assessment. Moreover, in the sub-criteria aggregation level using Choquet and Sugeno integrals, we have simply used a weighted sum which does not reflect the interactions among sub-criteria and the weights should be specified by the designer in addition to the fuzzy measures. However, the presented approach is suggested for a future work in which it can be used to build upon an interactive expert system (e.g., web-based) to gradually learn and optimize the measures for various occasions by clustering and analyzing the input data gathered for various design cases. This way, the precision of the fuzzy measures and consequently the multi-criteria design evaluation, can be improved by incorporating a wider and more correlated range of designers and industrial specialists. Consequently, after a period of time, the users will not be required to provide all the measures for each design case. Just providing their customization parameters and preferences would certainly suffice. Further studies should be also carried out towards the improvement of the methods of assessment for each criterion and sub-criterion and also validation of these methods especially in the cases of design complexity and flexibility. Ideally, as a future study, two cases of sensitivity analysis should be carried out to check the robustness of the global scores with respect to the variations of criteria values and also the effect of the lattice of fuzzy measures, i.e., μ_i, μ_{ij}, μ_{ijk}, … on the aggregation results. The latter would help us assess the sensitivity of the selection process with respect to the variations in designer's preferences.

Appendix

Multicriteria design of a mechatronic system: a questionnaire Q-ID:MMP016-2014

Consider the case of designing a mechatronic system such as a Quadrotor drone (as shown in Fig. 15).

The designer team has already come up with a set of five criteria as follows:

1. Increasing machine intelligence:
   including two components of control intelligence and interface intelligence.
2. Increasing system reliability.
3. Reducing design complexity:
   depends on the quantity of components, the degree of architecture complexity, number of feedback loops in design process, the number of distinct knowledge bases, the controller complexity and finally the extent of embedded software in product.
4. Increasing design flexibility:
   depends on the number of alternative component design paths, the number of components customization options and the number of choices for system architecture.
5. Reducing cost:



Fig. 15 Examples of mechatronic systems for a case of design

including cost of various parts, components, production, manufacture, etc.

Question 1) from 1 to 10, what do you intuitively think it is the degree of importance for each of the aforementioned 5 criteria in designing a good mechatronic product?

$\mu_1 =$	$\mu_2 =$	$\mu_3 =$	$\mu_4 =$	$\mu_5 =$

Question 2) from -10 to 10, what do you intuitively think it is the degree of correlation between each pair of criteria? What is the effect of increasing criterion i on criterion j? (Negative value represents negative correlation)

$m_{12} =$	$m_{13} =$	$m_{14} =$	$m_{15} =$
$m_{23} =$	$m_{24} =$	$m_{25} =$	
$m_{34} =$	$m_{35} =$		
$m_{45} =$			

Thank you for your participation

☺

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